EIT

EIT
Resourceful Links
https://www.ox.ac.uk/admissions/graduate/fees-and-funding/funding/under-represented-groups
https://www.eitcdt.ox.ac.uk/foaicdt
https://aims.robots.ox.ac.uk/
https://eng.ox.ac.uk/study/postgraduate/courses
https://www.robots.ox.ac.uk/~vedaldi/vacancies.html
https://www.robots.ox.ac.uk/~vgg/projects/visualai/
Oxford VGG https://www.robots.ox.ac.uk/~vgg/people.html
Unsupervised Vision (Vedaldi):
https://www.robots.ox.ac.uk/~vedaldi/research/union/union.html

Alice https://ibme.ox.ac.uk/person/alison-noble/?utm_source=chatgpt.com - Biomedical
Torr Group: https://www.robots.ox.ac.uk/~phst/

EIT Docs:
https://docs.google.com/document/d/1CkKFqahRNprK4I_ijlmxwOr4MKjnQHZ-sJorB4qVslM/edit?tab=t.rqbitvjcaml4
Application guide:
https://public.openwatercdn.com/1e1defaf-ff75-4676-8c8f-68cd36ddde6d/8cfbd37d-c2c6-41f5-bcb2-4902c64e9dac.pdf
Chat: https://chatgpt.com/c/692481ed-ea1c-8330-bcd6-e2fa04c6bfd2
EIT Review: https://gemini.google.com/app/569ac66e52a03a47
https://scholar.google.com/citations?hl=en&user=ED4V7hEAAAAJ&view_op=list_works&sortby=pubdate
Alison:
https://scholar.google.com/citations?hl=en&user=b0tmmYMAAAAJ&view_op=list_works&sortby=pubdate
https://www.ludwig.ox.ac.uk/research/jens-rittscher-group-page/jens-rittscher-group-page/research-overview
https://www.ludwig.ox.ac.uk/research/jens-rittscher-group-page/jens-rittscher-group-page/for-non-scientists

Graduate Funding | Ellison Institute of Technology

Oxford8cfbd37d-c2c6-41f5-bcb2-4902c64e9dac.pdf - Application guide

scholars@eit.org.

A-Z of postgraduate courses | Graduate Admissions | University of Oxford

International qualifications | University of Oxford

English language requirements | University of Oxford

MCML Application

MCML Application

- [ ] Identify the Professors I want to work with and the topics I would like to collaborate on with them.
- [ ] Be clear about the overall topics I would like to work on
- [ ] Motivation statement 250 words
- [ ] Brief Research Statement
- [ ] Select one of MCML’s research areas
- [ ] Academic CV Revised
- [x] ~~Complete Background section: Scientific Publications, Internships, Awards/Scholarships/Fellowships.~~
- [ ] Research Areas & Specific Topics
- [x] ~~References & Miscellaneous~~
- [x] ~~Transcript including conversion~~

Details on internships

For the free-text box asking for "relevant internships or professional work experience," you can list the following entries from your CV. I have formatted them to match the application's request.

- **Company:** Xulab, Carnegie Mellon University
  - **Position Title:** Research Intern
  - **Duration:** 5 months (August 2024 - Dec 2024)
  - **Key Tasks:** Developed a two-stage Cryo-ET pipeline for tomogram denoising and automated particle picking. Trained and integrated deep learning models (3D U-Net and DeepETPicker) into a unified workflow. Applied and validated the pipeline on public EMPIAR datasets.
- **Company:** Carnegie Mellon University Africa
  - **Position Title:** Research Associate
  - **Duration:** 6 months (June 2025 - Present)
  - **Key Tasks:** Focused on understanding of complex urban traffic scenes using vision foundation models (VFMs). Working on data curation pipelines with diffusion models for dataset augmentation and visual grounding in low-resource settings.
- **Company:** Carnegie Mellon University Africa
  - **Position Title:** Graduate Teaching Assistant
  - **Duration:** 9 months (Sept 2024 - May 2025)
  - **Key Tasks:** Designed and led recitations for Mathematical Foundations of ML and Applied Computer Vision. Developed supplementary materials to enhance student learning.

Valedictorian, Carnegie Mellon University Africa

- Graduated top of the M.S. Engineering AI class of 2025.

Smart Africa Scholarship Award

- A prestigious merit scholarship fostering digital transformation and leadership in Africa, with a focus on AI and technology innovation.

Jasiri Entrepreneurship Fellowship

- A selective fellowship supporting individuals creating high-impact ventures in Africa through entrepreneurial training and mentorship.

Local Chair, African Computer Vision Summer School (ACVSS) 2025

- Spearheaded local planning and institutional coordination for a premier 10-day event uniting top African talent with leading international researchers.

Specific research topics of interest:

Are there any particular topics... that you find particularly interesting?

Yes, my research interest is focused on developing **Foundation Models** that move beyond static pattern recognition to achieve a deep, causal understanding of the physical world.

My research interests are centred around 3 broad topics. First, I am interested in developing Truly **Perceptive Models**, which prioritize visual and geometric evidence rather than relying heavily on linguistic priors that may lead to inaccuracies or hallucinations. Second, my goal is to enhance **dynamic and temporal understanding** by designing models that learn explicit temporal and causal relationships. This includes building representations for dynamic data types, such as video, enabling the creation of "world models" that comprehend how scenes evolve over time, rather than simply recognizing their static components. My third interest is in developing **Causal & Explainable AI**, based on the idea that a model must be able to understand causal dynamics in its input for it to provide explainable outputs. I am also fascinated by the application of these techniques to scientific discovery, particularly in medical and biological fields.

My goal of building **causal and explainable AI** aligns directly with the research of **Prof. Zeynep Akata** on explainability and **Prof. Niki Kilbertus** on causal discovery in dynamic environments, including applications in biology and medicine. This is complemented by my interest in **dynamic and temporal understanding**, which connects to the foundational 3D vision and SLAM research from **Prof. Daniel Cremers'** lab, and my focus on **perceptive-first models**, which resonates with the generative vision work of **Prof. Björn Ommer**. My interest in the intersection of foundation models and biology aligns with the work led by **Prof. Julia Schnabel**.

Ultimately, my background in 3D computational biology imaging has given me a strong appreciation for how these foundational models can be applied to critical scientific challenges, such as the work led by **Prof. Julia Schnabel**.

My research interests are centered on three broad, interconnected topics. First, I am interested in developing 'Truly Perceptive Models,' which prioritize visual and geometric evidence rather than relying heavily on linguistic priors that may lead to inaccuracies or hallucinations. Second, my goal is to enhance dynamic and temporal understanding by designing models that learn explicit temporal and causal relationships. This includes building representations for dynamic data, such as video, enabling "world models" that comprehend how scenes evolve, rather than simply recognizing static components. My third interest is in developing Causal & Explainable AI, based on the conviction that a model must understand the causal dynamics in its input to provide explainable outputs. I am also fascinated by the application of these techniques to scientific discovery, particularly in the fields of medicine and biology.

My goal of building causal and explainable AI connects directly to the research of Prof. Zeynep Akata on explainability and Prof. Niki Kilbertus on causal discovery in dynamic environments. My second theme is connected to the work on foundational 3D vision and SLAM research from Prof. Daniel Cremers' lab. Similarly, my focus on 'perceptive-first' models aligns with the generative vision work of Prof. Björn Ommer. Finally, the MCML/Helmholtz environment is ideal, as it enables this foundational work to address critical applications; my interest in scientific discovery aligns perfectly with the research led by Prof. Julia Schnabel on foundation models for biomedical imaging.

MILA/EIT Todo

MILA (Supporting Documents - Matching M.Sc - MyMila | MyMila)

- [ ] Add academic Background
- [ ] Add publications
- [ ] Add coding experience
- [ ] Highlights and additional information
- [ ] Add Faculty Advisor and research interests

https://mila.quebec/en/directory/david-scott-krueger,
https://scholar.google.ca/citations?hl=en&user=5Uz70IoAAAAJ&view_op=list_works&sortby=pubdate
https://scholar.google.ca/citations?hl=en&user=XpscC-EAAAAJ&view_op=list_works&sortby=pubdate

MILA Supporting docs to prep

- [ ] CV
- [ ] Motivation/presentation letter (PDF, maximum three pages)
- [ ] Undergraduate transcripts PDF
- [ ] Master transcripts
- [ ] Relevant / selected works.

EIT(Ellison Scholars - Submission)

- [ ] Research Proposal
- [ ] Portfolio: In this section, you are asked to submit evidence demonstrating how well you align with the EIT vision. You may submit up to ten examples.

Referees & Contacts

Bosho CMU Referees & Contacts

Name	Position	Institution	Contact
Prof. João Barros	Associate Director and Research Professor, CMU-Africa Courtesy Appointment, Electrical and Computer Engineering	CMU Africa Carnegie Mellon University Africa Professor	jbarros@andrew.cmu.edu D206 Regional ICT Center of Excellence Bldg
Prof. Moise Busogi	Assistant Teaching Professor	CMU-Africa	mbusogi@andrew.cmu.edu
Prof. George Okeyo	Associate Teaching Professor and Director of Academics,	CMU-Africa	gokeyo@andrew.cmu.edu
Prof. Timothy Brown	Professor, Engineering and Public Policy Director, Kigali Collaborative Research Center	CMU-Africa	timxb@cmu.edu D206 Regional ICT Center of Excellence Bldg

I took Pro Moise's courses and worked as a teaching assistant for his courses.

Hi Prof, thanks again for offering to write recommendations for my PhD Applications. I am currently applying for the CMU Portugal Dual Degree PhD program (ECE), and I would really appreciate it if you could submit a recommendation letter for me.

MILA Links

My MILA Options:
Guy Wolf: https://guywolf.org/

https://cim.mcgill.ca/~pvg/people/
https://cim.mcgill.ca/~pvg/assets/positions/PVG_PhDAnnouncement_Version6.pdf
https://neo-x.github.io/join/#:~:text=Here%20is%20a%20mutable%20list,topics%20of%20interest%20when%20applying
https://cim.mcgill.ca/~pvg/assets/positions/PVG_PhDAnnouncement_Version6.pdf
file:///C:/Users/Bosho/Downloads/PhD%20Supervisor%20Search.pdf
https://chatgpt.com/c/691ee9d0-4d94-832a-95d4-7d7ae78d002f
https://aaroncourville.wordpress.com/
https://gemini.google.com/u/1/app/e42ab673fd023b9d?pageId=none

https://coursecatalogue.mcgill.ca/en/
https://admission.umontreal.ca/en/admissions/preparing-your-application/respect-official-deadlines/?gad_source=1&cHash=de4f01423b6d41667eed601232ba6789
https://www.mcgill.ca/gradapplicants/programs
https://www.hec.ca/en/programs/index.html
https://portal.mila.quebec/site/me/welcome
https://mila.quebec/en/prospective-students-and-postdocs/research-programs/request-supervisor
https://portal.mila.quebec/site/me/welcome
https://portal.mila.quebec/site/forms/matching-msc
https://mila.quebec/en/directory/simon-lacoste-julien
https://www.iro.umontreal.ca/~slacoste/
https://www.irina-rish.com/
https://schhttps://sites.google.com/view/mila-ood-rg/olar.google.com/citations?hl=en&user=Avse5gIAAAAJ&view_op=list_works&sortby=pubdate
https://mila.quebec/en/directory/simon-lacoste-julien
https://www.polymtl.ca/programmes/programmes/doctorat-en-genie-informatique
https://www.polymtl.ca/futur/es/programmes
https://www-labs.iro.umontreal.ca/~milacand/

Publication/Thesis

Paper Under review

Paper Title

FedProp: Federated Graph Neural Networks with Feature Propagation

Author

Brian Kipkirui

Status

Under Peer Review for a Major AI Conference

Abstract

Graph Neural Networks (GNNs) are a powerful tool to learn from relational data; however, training them in federated, privacy-preserving settings still poses considerable challenges. A key obstacle is the presence of so-called cross-client edges, i.e., edges whose incident nodes reside on distinct devices. Because standard message-passing GNNs aggregate features across neighboring nodes, these cross-client edges lead to information gaps that degrade accuracy when communication between client devices is disallowed. We recast this obstacle as a missing-feature problem and introduce FedProp: a model-agnostic federated GNN framework that reconstructs the l-hop features required for aggregation via feature propagation. In contrast to prior approaches, which either learn additional models to generate the missing neighbors or communicate the missing information, FedProp offers a lightweight iterative process that runs locally, requiring no inter-client communication or additional parameters. First, we provide a theoretical justification for feature propagation in subgraphs and provide a convergence guarantee of the iterative process. We then empirically demonstrate that FedProp can significantly improve GNN performance. For example, on Cora (non-IID) with a GCN backbone, FedProp achieves 81.0% accuracy, a 19.1% relative improvement over the federated baseline (0.680). FedProp thus enables users to deploy powerful GNNs across various client devices with locally stored data, achieving near-optimal accuracy and minimal communication costs..

Full Paper for Committee Review

The full research paper is provided below. It details the complete research process, including the conception of the algorithm from first principles, mathematical analysis with convergence guarantees based on spectral graph theory, and the implementation, accompanied by experimental results.

You can access the full PDF via the following secure link [here]

Todo (Focus on ELLIS)

Cold Email Candidates:
https://mcml.ai/research/groups/wiestler/ (AI for Image-Guided Diagnosis and Therapy at TU Munich),
https://scholar.google.de/citations?hl=en&user=-EEm-_MAAAAJ&view_op=list_works&sortby=pubdate , https://ai-idt.github.io/people/

Fabian MCML: https://mcml.ai/research/groups/theis/

MCML Call for applications: https://mcml.ai/opportunities/for-phd-applicants/mm2026-pis/
EIT Professors: https://www.eit.org/scholars/supervisors
https://ai4biomedicine.org/team/#open-positions

Todo (Focus on **ELLIS**)

- [ ] Work on the website (update the projects)
- [ ] Find Ellis Professors I am interested in
- [ ] Make contact with the labs
- [ ] Prepare application documents

Upcoming deadlines

- UvA Perceptual Systems (10th)
- Ellis central applications (31st)
- Tubingen Geiger ATG lab https://uni-tuebingen.de/en/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/team/application-guidelines/ (No deadline, although they take through ELLIS)
- Impris (November 16th): https://imprs.is.mpg.de/application
- Max Planck CLS/ETH: https://learning-systems.org/apply Nov 6
- ETH https://switzerland.recruitee.com/o/eth-ai-center-doctoral-fellowship-2026 Nov 7
- UTS: https://www.uts.edu.au/for-students/admissions-entry/how-to-apply/masters-by-research-phd OCT 29

CMU Portugal options:

https://scholar.google.com/citations?hl=pt-PT&user=WLOWIngoRCoC&view_op=list_works&sortby=pubdate

- https://web.fe.up.pt/~jsc/

Central Applications Vs Lab Applications vs. Scholarship Applications
Deadlines for each

Outreach packets include a personal statement and a research interest statement. Create a set of questions to help complete the latter.
Determine what to do for the applications this week.

Links:
Main Page: https://ellis.eu/news/ellis-phd-program-call-for-applications-2025

ELLIS Testimnonials.
https://ellis.eu/websites/european-lab-for-learning-intelligent-systems/pages/what-it-s-like-to-be-an-ellis-phd-student

Here's a **comprehensive todo list** for your ELLIS PhD application, organized in clear sections:

Section 1: Preparation & Research

- [ ] Review the official ELLIS PhD program page and FAQs.
- [ ] Mark key dates:
    - [ ] Info session: October 1, 2025.
    - [ ] Application portal opens: October 1, 2025.
    - [ ] Application deadline: October 31, 2025.
- [ ] Register for the October 1st virtual info session (recommended).
- [ ] Research participating advisors and prepare a shortlist.
- [ ] Identify your desired track (Academic / Industry / Interdisciplinary).

Section 2: Document Preparation

- Write a two-page motivational letter (statement of purpose + research statement).
- Update your academic CV with full citations, presentations, and awards.
- Gather unofficial transcripts for BSc and MSc degrees (with English translation).
- Gather optional materials:
    - Published paper, master's thesis, or portfolio as a single PDF (<20 MB).

Section 3: Advisor & Track Selection

- Explore the list of available advisors when portal opens.
- Select your preferred research fields and potential supervisors.
- Double-check if parallel applications are required for your preferred advisor's institution (submit if needed).

Section 4: Reference Management

- Identify at least 2-3 academic referees suited to write strong recommendation letters (ideally including your MSc thesis supervisor or PI).
- Contact referees well before the deadline and secure their agreement.
- Prepare referee contact details to enter into the portal.
- Send formal reference requests via the portal as soon as it opens.
- Track referee submission progress; send reminders if needed (before Oct 31).

Section 5: Portal Application

- Register on the ELLIS application portal (after October 1, 2025).
- Complete the application form thoroughly with accurate personal, academic, and experience details.
- Upload all necessary documents as PDFs.
- Indicate your preferences for field/advisors.
- Double-check and submit your application before October 31, 2025, 23:59 AoE.

Section 6: After Submission

- Confirm receipt of reference letters (by November 25).
- Monitor email for info on interview stage (Jan/Feb 2026) and further updates.
- Prepare for possible interviews (January-February 2026), reviewing your proposal, background, and motivation.

Section 7: Miscellaneous/Follow-up

- Subscribe to the ELLIS newsletter if interested.
- Follow up with your referees as needed.
- Prepare any parallel university applications if necessary due to your advisor's institutional requirements.
- Save all final documents/submissions for your records.

Let me know if you'd like this list as a downloadable checklist, a Notion page, or a Todoist/Trello import.

1. https://ellis.eu/news/ellis-phd-program-call-for-applications-2025

Tab 2

1. Theory vs. Empiricism

This dimension measures your preference for mathematical proofs and guarantees versus data-driven discovery.

- **Theoretical Contributions: 4.5 / 5.0**
- **Empirical Investigations: 3.5 / 5.0**

Interpretation: You have a strong preference for theoretically-grounded work but are also motivated by powerful empirical discoveries.

2. Scale vs. Efficiency

This dimension contrasts the appeal of massive, powerful models with that of clever, resource-efficient algorithms.

- **Efficiency-Focused Research: 4.0 / 5.0**
- **Scale-Focused Research: 2.0 / 5.0**

Interpretation: This is your most decisive dimension. You show a very strong preference for efficiency and clever algorithms over a "brute force" or scale-based approach.

3. Interpretability vs. Performance

This dimension weighs the importance of understanding a model's decisions against achieving maximum predictive accuracy.

- **Interpretability-Focused Research: 5.0 / 5.0**
- **Performance-Focused Research: 5.0 / 5.0**

Interpretation: You have a rare and powerful dual interest. You are equally and maximally motivated by both making AI transparent and pushing it to the absolute limits of its performance capabilities.

4. Interdisciplinary vs. Core ML

This dimension measures your interest in applying ML to other fields versus advancing the fundamental methods of ML itself.

- **Core ML Advancement: 4.5 / 5.0**
- **Interdisciplinary Application: 3.3 / 5.0**

Interpretation: You have a clear and strong inclination towards advancing the core algorithms, systems, and capabilities of machine learning.

5. Societal Impact vs. Technical Purity

This dimension contrasts research focused on broad societal/ethical issues with research focused on the technical challenges of making AI safe and robust.

- **Technical Purity & Safety: 4.25 / 5.0**
- **Broad Societal & Ethical Impact: 3.1 / 5.0**

Interpretation: Your interest lies more in the technical, algorithmic challenges of making AI robust, safe, and reliable, rather than its broader societal applications or ethical considerations.

Prompt:

PhD Lab Search for a Foundational Computer Vision Researcher

Objective:

Your task is to identify a prioritized shortlist of 20 leading professors and their labs within the ELLIS network who are at the forefront of foundational computer vision research. The ideal candidate is a dedicated computer vision researcher who seeks to build the next generation of vision systems, guided by a strong preference for theoretical depth, algorithmic efficiency, and interpretability, often with applications in science.

Primary Search Source:

Focus exclusively on the official ELLIS network website (ellis.eu). Systematically search the directories of ELLIS Fellows, Scholars, and the descriptions of ELLIS Units.

Core Research Profile to Match:

1. Non-Negotiable Foundation: Computer Vision Excellence

The primary and essential criterion is that the lab's central focus must be Computer Vision. The professor should be a recognized leader in the field, regularly publishing in top-tier vision conferences such as CVPR, ICCV, and ECCV.

2. Secondary Filters (The Lab's Philosophical "Flavor"):

After confirming a lab is a top-tier vision lab, screen it for the following characteristics. The best matches will exhibit a strong combination of these traits within the context of their vision research:

- **Theoretical & Principled:** The lab doesn't just build models, it seeks to understand the mathematical principles behind them. (e.g., *Why* do certain architectures work? What are the geometric properties of visual data?).
- **Efficient & Resourceful:** The lab develops vision models that are fast, compact, and can run under constraints (e.g., on mobile devices or for real-time video).
- **Interpretable & Causal:** The lab works on making "black box" vision models transparent, explaining what they are looking at and why they make certain decisions.
- **Application in Scientific Imaging (High-Value Bonus):** The lab applies its novel vision methods to solve complex scientific problems (e.g., analyzing medical scans, microscopy, or satellite data). This demonstrates an interest beyond standard vision benchmarks.

Keyword Search & Screening Strategy:

Step 1: Identify Top Computer Vision Labs

Scan all profiles for primary keywords indicating a core focus on Computer Vision.

- **Primary Keywords:** Computer Vision, 3D Vision, Visual Recognition, Image Understanding, Video Analysis, Computational Photography, Scene Reconstruction.

Step 2: Apply Philosophical Filters

From the list of vision labs, prioritize those whose research descriptions and publication lists also contain keywords related to your specific interests:

- **For Theory:** Geometric Deep Learning, 3D Reconstruction Theory, Representation Learning, Mathematical Foundations of Vision.
- **For Efficiency:** Efficient Computer Vision, Model Compression, Quantization, Real-Time Vision, Video Compression.
- **For Interpretability:** Explainable AI (XAI) for Vision, Interpretable Vision Systems, Visual Attribution, Causal Reasoning in Vision.
- **For Scientific Application:** Medical Image Analysis, Computational Pathology, Digital Pathology, Satellite Image Analysis, Computational Microscopy, Bioinformatics (if vision-related).

Step 3: Final Output

Produce a ranked list of the top 20 professors/labs. For each entry, provide the following:

- **Professor Name:**
- **University & ELLIS Unit:**
- **Evidence of Vision Leadership:** Note their publications in CVPR, ICCV, ECCV or leadership roles in the vision community.
- **Alignment with Your Profile (Justification):** In 2-3 sentences, explain how this vision lab specifically aligns with your philosophical interests. For example: "Professor X is a leader in 3D Vision (CVPR Best Paper). Their lab aligns perfectly by focusing on the *theoretical principles of geometric deep learning* and applying these efficient new models to the challenge of *interpreting 3D medical scans*."

ShortListed Profs

Gemini: https://gemini.google.com/u/1/app/fcfc9a2829350ad2
GPT Report: https://chatgpt.com/c/68c959d9-0044-8329-b525-ff00a3df9e3e ,
file:///Users/brianbosho/Downloads/PhD%20lab%20search.pdf

Excellent. Upping the target to 20 provides a much wider net and gives you more options to explore. Based on an extensive search of the ELLIS network, leveraging the precise, computer-vision-centric prompt we designed, here is a list of 20 professors and labs that align with your unique profile.

They are grouped thematically to help you see the different "flavors" of research that match your interests.

Category 1: The 3D Vision & Geometric Deep Learning Pioneers

(These labs are at the absolute core of your interest in the theoretical and geometric foundations of vision.)

1. **Daniel Cremers** (ELLIS Unit Munich, TU Munich)
    - **Evidence of Vision Leadership:** A world-renowned leader in 3D vision, computer graphics, and autonomous systems. Recipient of the Leibniz Prize. Numerous papers at CVPR, ICCV, ECCV.
    - **Alignment:** This is a top-tier choice. His lab combines deep theoretical and mathematical rigor (variational methods, geometric deep learning) with cutting-edge applications in 3D reconstruction, SLAM, and autonomous navigation.
2. **Michael M. Bronstein** (ELLIS Unit Cambridge, University of Cambridge)
    - **Evidence of Vision Leadership:** A key figure in the field of geometric deep learning, particularly on graphs and manifolds.
    - **Alignment:** If you are interested in the fundamental mathematical principles behind deep learning for vision, this is a prime lab. His work on Graph Neural Networks (GNNs) is foundational and has wide-ranging applications, from 3D vision to scientific problems.
3. **Matthias Nießner** (ELLIS Unit Munich, TU Munich)
    - **Evidence of Vision Leadership:** Leading expert in 3D reconstruction, semantic scene understanding, and video manipulation. Multiple best paper awards at top conferences.
    - **Alignment:** His lab is at the forefront of combining classical geometry with modern deep learning for 3D vision. The focus on real-time, efficient 3D capture and understanding aligns perfectly with your interest in efficiency.
4. **Vladlen Koltun** (ELLIS Unit Tübingen, Apple)
    - **Evidence of Vision Leadership:** Highly influential researcher with a history at Intel and now Apple. Known for high-impact work in 3D vision, robotics, and computational photography.

- **Alignment:** His research embodies the blend of fundamental research with real-world impact. He has a strong track record of tackling hard, foundational problems in vision with an eye toward creating robust and usable systems.

5. **Andrea Vedaldi** (ELLIS Unit Oxford, University of Oxford)
   - **Evidence of Vision Leadership:** A leading name in visual recognition, deep learning theory, and geometric methods in vision. Co-author of the popular VLFeat library.
   - **Alignment:** His group focuses on understanding the "why" behind deep learning in vision. This aligns perfectly with your interest in theory and interpretability, with a strong foundation in core vision problems.

Category 2: The Medical & Scientific Vision Specialists

(These labs are top-tier vision labs that use their expertise to solve hard problems in science, a key bonus for you.)

6. **Carola-Bibiane Schönlieb** (ELLIS Unit Cambridge, University of Cambridge)
   - **Evidence of Vision Leadership:** Head of the Cambridge Image Analysis group, expert in the mathematics of imaging.
   - **Alignment:** A perfect match for your interest in the intersection of theory and scientific application. Her lab develops new mathematical and ML models for inverse problems in imaging, with a heavy focus on biomedical and clinical applications.
7. **Daniel Rueckert** (ELLIS Unit Munich, TU Munich & Imperial College London)
   - **Evidence of Vision Leadership:** A world leader in medical image analysis and AI in healthcare.
   - **Alignment:** This lab is a premier example of applying state-of-the-art computer vision to medicine. They work on the interpretation of complex medical scans (MRI, CT), aligning perfectly with your interest in a high-impact scientific application of vision.
8. **Nassir Navab** (ELLIS Unit Munich, TU Munich)
   - **Evidence of Vision Leadership:** Pioneer in medical augmented reality and machine learning for computer-aided medical procedures.
   - **Alignment:** This lab blends core computer vision research with direct clinical applications. It's a great fit if you're interested in how vision can augment human experts in high-stakes environments like surgery.
9. **Julia Schnabel** (ELLIS Unit Munich, TU Munich & Helmholtz Center)
   - **Evidence of Vision Leadership:** Expert in computational imaging and analysis, particularly for cancer imaging.
   - **Alignment:** This is a strong, science-driven vision lab. Their work on creating intelligent, efficient, and interpretable models for medical imaging (e.g., pathology, radiology) is a direct hit on your stated interests.

Category 3: The Efficiency & Interpretability Champions in Vision

(These labs focus on making vision systems faster, smaller, more transparent, and more reliable.)

10. **Bernt Schiele** (ELLIS Unit Tübingen, MPI for Informatics)
    - **Evidence of Vision Leadership:** A highly respected, long-time leader in computer vision, focusing on object recognition, tracking, and scene understanding.
    - **Alignment:** His lab has a strong focus on robustness, fairness, and understanding the limitations of current vision models. This aligns perfectly with your interest in the theoretical and interpretability aspects of vision.
11. **Jiri Matas** (ELLIS Unit Prague, Czech Technical University)
    - **Evidence of Vision Leadership:** A giant in the field, known for his seminal work on object recognition, tracking, and image matching (e.g., MSER detector).
    - **Alignment:** His group is known for producing efficient and robust algorithms that become industry standards. This is a great fit for your interest in creating foundational, efficient vision technology.
12. **Bastian Leibe** (ELLIS Unit Aachen, RWTH Aachen University)
    - **Evidence of Vision Leadership:** Leading researcher in object recognition, tracking, and perception for robotics and autonomous systems.
    - **Alignment:** His lab combines core vision research with the demands of real-world robotic systems, which necessitates a focus on efficiency and robustness. This is a great practical application of your core interests.
13. **Laura Leal-Taixé** (ELLIS Unit Munich, TU Munich)
    - **Evidence of Vision Leadership:** Expert in video analysis, multi-object tracking, and understanding dynamic scenes.
    - **Alignment:** Her work on tracking and video understanding requires efficient models. The lab also works on interpretability, trying to understand the "why" behind model predictions in complex, crowded scenes.

Category 4: Vision at the Intersection of Other Core ML Fields

(These labs have a strong vision focus but also incorporate elements from other areas like generative models, NLP, and robotics.)

14. **Cordelia Schmid** (ELLIS Unit Paris, Inria)
    - **Evidence of Vision Leadership:** A world-class leader in computer vision, particularly in action recognition, large-scale retrieval, and vision+language models.
    - **Alignment:** Her lab is at the forefront of large-scale vision. While this might seem to run counter to "efficiency," their work on representation learning is foundational and touches on many of your core interests.

15. **Josef Sivic** (ELLIS Unit Prague, CIIRC CTU Prague)
    - **Evidence of Vision Leadership:** Leading researcher in large-scale visual search, object recognition, and linking vision with text.
    - **Alignment:** His work on understanding scenes from images and video, often with weak supervision, requires clever and efficient methods. This is a great lab for foundational work on visual understanding.
16. **Tin_e Tom_a__zi_c** (ELLIS Unit Ljubljana, University of Ljubljana)
    - **Evidence of Vision Leadership:** Well-known for his work in visual tracking and recognition.
    - **Alignment:** This lab has a strong track record of producing efficient and robust algorithms for core vision tasks. A good fit for a focus on practical, high-performance vision systems.
17. **Luc Van Gool** (ELLIS Unit Zurich, ETH Zurich)
    - **Evidence of Vision Leadership:** One of the most prolific and influential researchers in European computer vision. His work spans nearly every area of the field.
    - **Alignment:** Given the sheer breadth of his lab's work, you would almost certainly find a project that aligns with your interests, from efficient vision to 3D reconstruction and scientific applications.
18. **Theo Gevers** (ELLIS Unit Amsterdam, University of Amsterdam)
    - **Evidence of Vision Leadership:** Expert in color, object recognition, and human-centric computer vision.
    - **Alignment:** His lab often focuses on the "why" of perception, including affective computing and interpretability. A good fit for your interest in the more human-centric and explanatory side of vision.
19. **Nicu Sebe** (ELLIS Unit Trento, University of Trento)
    - **Evidence of Vision Leadership:** Focuses on human behavior understanding, multimedia analysis, and affective computing.
    - **Alignment:** Similar to Theo Gevers, this lab offers a path to work on the interpretability and human-centric aspects of computer vision, particularly in understanding social signals and emotions.
20. **Serge Belongie** (ELLIS Unit Copenhagen, University of Copenhagen)
    - **Evidence of Vision Leadership:** A highly respected figure known for his work in object recognition, image segmentation, and pioneering work in fine-grained categorization.
    - **Alignment:** His research has often blended core vision problems with applications that require a deep understanding of visual data (e.g., biodiversity). This aligns with your interest in science-driven vision problems that require more than just a label.

CLS Slides

1. Overall structure (aim for ~10–12 minutes, not full 15)

I'd structure it like this:

1. **Intro & Background** – 1–1.5 min
2. **Unifying Research Theme** – 1 min
3. **Project 1: Cryo-ET (robust perception under noise)** – 2 min
4. **Project 2: FedProp (theory + efficiency under privacy/constraints)** – 3–4 min
5. **Project 3: Urban Mobility & Foundation Models** – 2–3 min
6. **Future Research Aims (your PhD vision)** – 2–3 min
7. **Closing (why CLS / who you want to become)** – 30–60 sec

Total: ~11–13 minutes. That's perfect: under 15, but substantial.

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2. Slide-by-slide outline

Slide 1 – Title & Intro

Title:
"Towards Robust, Dynamic, and Causally-Grounded Perception for Foundation Models"

Content:

- Name, affiliation (CMU-Africa, Research Associate)
- Degrees (MSc in Engineering AI)
- One-line research theme

What you say (roughly):

"Hello, my name is Brian Kipkirui. I'm a Research Associate at Carnegie Mellon University Africa, where I also completed my master's in Engineering Artificial Intelligence.

My research is driven by a simple conviction: if AI is to be useful in the physical world, it must move beyond static pattern recognition to robust, dynamic, and causally grounded perception. In this talk, I'll briefly introduce my background, focus on three research projects I've worked on, and then outline the PhD research directions I want to pursue at CLS."

Slide 2 – Educational Background

Content bullets:

- BSc in Electrical & Electronics / Electronics & Computer Eng (JKUAT)
- MSc Engineering AI, CMU-Africa (mention valedictorian if you want, but small font)
- Current position: Research Associate, CMU-Africa
- Core areas: computer vision, representation learning, federated learning, mobility

What you say:

"I did a dual bachelor's in Electrical & Electronics Engineering and Electronics & Computer Engineering, and then an MSc in Engineering Artificial Intelligence at CMU-Africa.

During my master's and now as a Research Associate, I've focused on three themes: robust perception from imperfect data, mathematically grounded algorithms, and foundation models in resource-constrained environments."

Slide 3 – Unifying Research Theme

Title: "Unifying Theme: Robust Perception Under Real-World Constraints"

Visual idea:
A simple diagram with three boxes:

- Noisy / weak labels (Cryo-ET)

- Privacy / limited communication (FedProp)
- Limited sensors / dynamic scenes (Mobility)

All pointing to: “Robust, dynamic, causally grounded models”.

What you say:

> “Across three projects, the core question has been the same:
> *how do we make models reliable when the world is noisy, constrained, and dynamic?*
>
> I’ll briefly discuss:
>
> - Cryo-ET: robustness under extreme noise and weak labels
> - FedProp: robustness and efficiency under privacy and communication constraints
> - Urban mobility: robustness of foundation models under limited sensing and high dynamics
>
> Then I’ll connect these experiences to the PhD agenda I propose.”

This sets the frame so they see *coherence*, not a random list.

Slide 4–5 – Project 1: Cryo-ET (3D noisy perception)

Slide 4 – Problem & Context

- Title: “3D Cryo-ET: Perception Under Extreme Noise”
- Bullet points:
  - 3D Cryo-ET tomograms: noisy, low SNR, weak labels
  - Goal: denoising + automated particle detection
  - Role: you built a deep-learning pipeline end-to-end

Slide 5 – Key Insight & Result

- Visuals:
    - A schematic: noisy 3D volume → denoiser → detector
    - Maybe a tiny example image (if you have)
- Key bullet:
    - Automation vs manual inspection
    - Main insight: robust perception = inferring structure from uncertainty

What you say (across the two slides):

> "At Xulab, I worked on 3D Cryo-Electron Tomography. The data are extremely noisy, weakly labeled 3D volumes, where you want to detect and analyze biological structures.
>
> I developed a deep-learning pipeline combining denoising and automated particle picking. This replaced a largely manual process and allowed researchers to inspect more structures more quickly.
>
> The main *research* insight for me was that standard models trained on clean images degrade heavily under this kind of noise. Robust perception in scientific settings means learning to infer 3D structure from uncertainty, not just recognizing patterns in curated datasets. This was my first deep encounter with perception as inference under imperfect information."

Don't overdo technical details here; emphasize *intuition* + *lesson*.

Slide 6–8 – Project 2: FedProp (theory-heavy)

You can afford to spend a bit more time here because it shows deep technical ability.

Slide 6 – Problem & Formulation

- Title: "FedProp: Federated GNNs Under Privacy and Communication Constraints"
- Diagram:
    - Several clients with local graphs → server

    - Missing neighbor links across clients = “missing neighbor problem”
- Bullets:
    - Federated GNNs: cannot share full graph due to privacy
    - Missing neighbor features degrade performance
    - Need: local propagation that approximates centralized solution

Slide 7 – Method & Theory

- Title: “Method: Local Feature Propagation with Theoretical Guarantees”
- Bullets:
    - Local propagation using graph Laplacian (no extra communication)
    - Spectral graph analysis of propagation operator
    - Convergence guarantees / error bounds vs centralized

You can include one simple equation or operator if you like, but keep it readable.

Slide 8 – Results & Insight

- Title: “Results & Takeaways”
- Bullet:
    - 99.6% of centralized performance on Cora
    - Zero additional inter-client communication
- Insight line:
    - “Strong math foundations → simple, efficient algorithms”

What you say:

> “In federated graph learning, privacy and communication costs introduce a different constraint. Clients each hold part of a global graph; they can’t share raw data. This

leads to what we call the ‘missing neighbor problem’: nodes whose neighbors live on other clients and whose features you never see.

In FedProp, I formulated this problem and designed a local feature propagation method that uses only each client’s local graph and a small number of propagation steps to approximate the centralized GNN.

The work involved applying spectral graph theory to analyze the propagation operator, deriving convergence guarantees and error bounds, and then validating them empirically. On Cora, FedProp achieved about 99.6% of centralized performance with zero additional inter-client communication.

For me, the deeper lesson was that strong mathematical foundations—linear algebra, spectral methods, optimization—are not abstract exercises. They are the tools that let you design *simple, efficient algorithms* instead of just stacking more parameters.”

Slide 9–10 – Project 3: Urban Mobility & Foundation Models

Slide 9 – Context & Constraints

- Title: “Urban Mobility in African Cities: Foundation Models Under Real Constraints”
- Bullets:
    - Low-resource setting: only monocular cameras, no fancy sensors
    - Goals: traffic understanding, safety, efficiency
    - Constraints: limited sensors, dynamic scenes, distribution shift

Slide 10 – FM Limitations & Experiments

- Bullets:
    - VLM/VFM issues: depth, occlusion, motion, OOD robustness
    - Using diffusion-based augmentation, geometry-aware cues, etc.
    - Societal impact: mobility is a daily problem for millions

What you say:

"Currently, I work on traffic understanding in African cities using camera data. In these settings, it's unrealistic to deploy expensive sensors like LiDAR; often, we have only a few cheap monocular cameras.

This makes the task a great stress-test for foundation models. We see them struggle with occlusion, depth, motion, and with scenes that differ from the distributions they've seen during pretraining.

I've been using generative models and vision foundation models—such as diffusion-based augmentation and geometry-aware cues—to improve robustness from just a single camera stream.

This work matters practically—mobility deeply affects quality of life in many developing countries—but it also sharpened my research vision: foundation models will only be useful in the physical world if they can operate under real constraints: limited sensors, dynamic environments, and imperfect data."

Slide 11–12 – Future Research Aims (CLS-focused)

You can do this as either:

- one slide with 3 columns (Aim 1/2/3), or
- two slides: one high-level, one connecting to CLS themes.

Slide 11 – Three Aims

Title: "PhD Vision: From Snapshots to Dynamic, Causal World Models"

Columns:

1. **3D-Grounded Perception**
   - 3D structure (depth, surfaces, spatial layout)
   - Monocular 3D cues, multi-view, or learned 3D priors
2. **Temporal State & Motion**
   - Compact latent state
   - Motion, interactions, temporal continuity

3. **Causal, Generative World Models**
    - Predict plausible futures
    - Answer “what if” counterfactuals for planning/control

What you say:

“These projects converge to a simple realization: today’s foundation models are very good observers of snapshots, but poor reasoners about change. They handle ‘what is’, but struggle with ‘what happened, why, and what will happen next if I act’.

In my PhD, I’d like to tackle this through three connected directions:

- First, building **3D-grounded multimodal perception** that understands depth and structure, not just 2D pixels.
- Second, giving these models a **compact temporal state**, so they can track motion and interactions over time.
- Third, developing **causal, generative world models** that can forecast futures and support counterfactual reasoning—essential for safe control.”

Slide 12 – Why CLS

Title: “Why CLS”

Bullets (keep short):

- ETH: 3D perception, robotics, systems
- MPI-IS: generative modeling, intelligent systems, theory
- Direct mapping to G2/G3/G4/G8 (you can keep these as small tags)

What you say:

“The Max Planck ETH Center for Learning Systems is the only environment I know that combines, so tightly, deep learning theory, 3D perception, and embodied systems.

ETH’s strengths in reliable perception and robotics, and MPI-IS’s expertise in generative modeling and intelligent systems, are exactly what my three aims require. I see my work as directly connected to CLS themes on robust perception, model-based control, perception-action loops, and multimodal foundation models."

Slide 13 – Conclusion & Future Self

Title: "Long-Term Vision"

Bullets:

- Become a researcher bridging theory + embodied systems
- Build capacity & mentorship in Africa
- Continue roles like ACVSS, collaborations, etc.

What you say:

> "Long-term, I want to be the kind of researcher who is rigorous in theory, grounded in real systems, and committed to building research capacity—especially in Africa.
>
> Organizing the African Computer Vision Summer School was one step in that direction, and I would like to continue building bridges between strong research hubs like CLS and emerging communities on the continent.
>
> A PhD at CLS is not just the next academic step for me; it's the environment I need to grow into that role. Thank you for your time and consideration."

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3. Recording & delivery tips

- **Aim for 10–12 minutes**, not 14: being crisp is a plus.
- Use **very visual slides**:
  - Small diagrams > lots of text.
  - One key number or equation per project is enough.
- Have your **motivation letter printed mentally** as backup structure, but don't read it.

- Practice **once with a timer**; adjust where you're too detailed.
- For Zoom recording:
  - Use screen share + camera (small box of you is nice).
  - Speak a tad slower than usual; committees often listen in batches.

If you want, next step I can:

- Turn this into a **bullet-point speaker notes document** you can put beside your slides, or
- Help you tighten wording for any specific slide once you draft them.

Labs & COntacts based on papers

Labs & Contacts based on papers

From Drive LM paper: Tubingen AVG Lab: https://www.katrinrenz.de/ , https://kashyap7x.github.io/ , https://uni-tuebingen.de/en/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/home/

Tubingen Applications:

A statement of research interests

A statement of research interests
(Considering the compositions we used to write this should be super easy as it can be rehearsed)

Cheat sheet from the professor:
Q1 Which topics in research are you most excited about?

Q2. Which areas of research do we most align on (i.e. a past paper of mine or yours).

Q3. What do you think are the most pressing challenges in AI/ML (think 3-5 year period). Tell me more about why you think you are uniquely interested/equipped to attack these challenges and how I can support your interests.

My Primary research interests are in Deep learning, machine learning, and computer vision. I am specifically interested in the use of self-supervised and weakly supervised approaches in computer vision to train foundation models for various tasks in AI for science. The main challenge in various scientific tasks from biology to other tasks is the ability to capture large amounts of data and the need for experts to annotate the data. Self supervised learning allows us to learn the general underlying peroperties and patterns in our data without any labels. Weakly supervised learning in the other hand allows us to learn from data and perform tasks where we have very few labels.

Recommendation letter

Recommendation letter

Dear Professor XXXX

I am writing as the Director of Academic Affairs at Carnegie Mellon University Africa to strongly recommend Btian Kipkirui for the [XXXX program] in your institution. I have known Brian since his first semester in my role overseeing academic affairs in the university, and also as an educational mentor. In this way, I have observed his academic performance and growth as a researcher and scholar at CMU Africa.

Brian distinguished himself as one of the top students in the Class of 2025, completing his Master of Science degree in Engineering with a focus on Artificial Intelligence (AI) with a near-perfect GPA of 3.97/4.0. Through the program, Brian took a large breadth of courses and excelled across both foundation and applied courses, illustrating both breadth and depth in mastery of different areas. Additionally, Brian went ahead to seek research opportunities, including taking a course as an independent study in applications of computer vision in computational biology, where he demonstrated his capabilities to work independently as a researcher and quickly master new fields. In this way, he demonstrated intellectual curiosity and independence engaging in activities beyond his coursework. His ability to resume graduate studies after more than 4 years gap since his undergraduate studies and rank at the very top of his class illustrates resilience, determination, and adaptability.

In addition to these, Brian also contributed substantially to the academic community in CMU-Africa through teaching assistance, club leadership, and mentorship. He served as a teaching assistant for mathematical foundations of machine learning where he offered regular office hours, and also prepared supplementary material for the course helping to shape the course structure and delivery. He also volunteered as a TA for a different course, Applied Computer Vision, going out of his way to offer support to students. In addition to that, Brian was a founding leader of the Africa Innovation society, an entrepreneurship club in CMU-Africa. Through his roles as a teaching assistant and club leader, brian helped mentor several students and enriched the experience of many students. These also illustrate his commitment to leadership, mentorship, and service.

In the course of my time in CMIU-Africa Brian stands out and is one of the very best students I have mentored. The combination of his determination, curiosity, brilliance, and his teaching and research abilities makes him an exceptional candidate for the doctoral program. I am confident that he will thrive through the PhD program, having shown persistence and independence. I recommend Biran without any reservation. He will not only successful in the program but will also enrich the research community. Please do not hesitate to contact me in case you need additional information.

Dear Prof XXX

I would like to recommend Brian for the position of XXXX in your school/institution.

I have known Brian in my position as the Director of Academic Affairs in CMU-Africa, where Brian has been a master's student.

Throughout the time Brian has been an exceptional student. He has consistently performed well in his courses and emerged as the top student in his graduating class in 2025.

Additionally, Brian has served as a teaching assistant for different courses in CMU-Africa, where he has shown a commitment to helping students in their coursework.

I strongly recommend Brian; I believe he would be a good fit for the doctoral position. If you need any additional information, feel free to contact me.

Best regards.

Useful facts

1) Completed courses covering a wide breadth: From fundamentals such as mathematical foundations of ML and Introduction to ML to more applied courses such as Applied Computer Vision and NLP
2) Despite being out of school for about five years after undergrad, I was able to pick up challenging courses, master the fundamentals, and still do well
3) Worked as a Teaching assistant
4) Known the student since the first semester as the director of student affairs and as a mentor to the student.
5) Student was the best in his graduating class and awarded the valedictorian award.

1. Academic Performance Context

- **How exceptional is “top student”?**
  Was Brian #1 out of how many? Did he maintain a GPA significantly higher than peers?

  He was the best ina graduating class of about 120 students, and maintained a GPA of 3.97

- **How broad was his mastery?**
  Did he only excel in applied courses, or also in foundational/theoretical ones? Any special recognition from faculty?
  - Had an A in all courses

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2. Intellectual Traits

- **How does Brian approach learning?**
  Curiosity, independence, persistence? Did he actively seek out difficult material or push beyond the curriculum?

  High degereee of curiosity, exploring extensive material for coursework and resources beyond the classroom. books….

- **How did he handle the gap after undergrad?**
  This shows resilience and adaptability — was there an observable difference in how he "caught up" and outpaced peers?

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3. Teaching & Mentorship

- **What was his impact as a TA?**
  Did students give feedback about his clarity, patience, or ability to explain concepts?
  - Offered regular office hours to support students and provided support in preparing supplementary material for students and shaping the course structure.. For mathematical foundations of ML

- **Did he go beyond routine TA duties?**
  For example, creating new teaching material, mentoring struggling students, or helping shape the course structure.
  - Also volunteered as a TA for a different course (Applied Computer Vision),

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4. Leadership & Character

- **How did you see him as a mentor or community member?**
  Beyond TA work, did he support peers, organize groups, or act as a role model?
  - Supported incoming students from the Kenyan comunity acting as a leader and mentor of Kenyan students in CMU-Africa.

- **What qualities stand out to you personally?**
  Reliability, professionalism, initiative, empathy, integrity?
  - Most of the above

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5. Comparative Assessment

- **Where does Brian rank among students you've overseen?**
  For example: "Among the top 1–2% of students I have encountered in my X years as Director."
    - He is one of the best students we have had in CMU-Africa over my 3 years in CMu.

- **Would you trust him with responsibilities beyond academics?**
  This tells a committee they can rely on him not just as a researcher, but as a community builder.
    - Definately

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6. Fit for Doctoral Studies

- **Why is he a good fit for a PhD program specifically?**
  Ability to handle independent work, persistence, ability to contribute to research culture.
    - All the above.

- **What unique strengths does he bring to a doctoral environment?**
  Breadth of knowledge, teaching experience, resilience, institutional leadership.
    - Has developed the right academic background from the courses he has had, he has had experience in research through courses he has done and working as a research associate in CMU over

Perfect — with those details, the Director’s letter can move from a short endorsement to a **powerful, distinctive recommendation**. Below is a polished draft that weaves in the facts you shared. It keeps the authoritative, institutional voice of a Director while giving admissions committees exactly what they want: clear evidence, comparative context, and a strong endorsement.

Draft Recommendation Letter

Dear Prof. [XXX],

I am writing in my capacity as Director of Academic Affairs at Carnegie Mellon University Africa to strongly recommend **Brian Kipkirui** for the [doctoral position/PhD program] at your institution. I have known Brian since his first semester as a master’s student, both as his academic mentor and in my role overseeing the academic affairs of the university. This vantage point has allowed me to observe his academic performance, his contributions to our community, and his growth as a scholar and leader.

Brian distinguished himself as the **top student in a graduating class of about 120 students**, completing his Master of Science in Engineering Artificial Intelligence with a near-perfect GPA of **3.97/4.0**. He earned an “A” grade in every course, excelling equally in rigorous theoretical foundations—such as *Mathematical Foundations of Machine Learning* and *Introduction to Machine Learning*—as well as in applied, project-based courses including *Applied Computer Vision* and *Natural Language Processing*. His record reflects not only raw intellectual ability, but also a breadth and depth of mastery that is rare.

Beyond formal coursework, Brian demonstrated **intellectual curiosity and independence** by engaging extensively with material beyond the classroom—consulting advanced texts, exploring supplementary resources, and seeking to connect theory with application. His ability to resume graduate studies after a five-year gap since his undergraduate degree, and then to rise to the very top of his cohort, is testimony to his adaptability, resilience, and determination.

Brian has also contributed substantially to our academic community through **teaching and mentorship**. He served as a Teaching Assistant for *Mathematical Foundations of Machine Learning*, where he not only offered regular office hours but also prepared supplementary material and supported the shaping of course structure. He further volunteered as a TA for *Applied Computer Vision*, demonstrating initiative and commitment well beyond his assigned responsibilities. His effectiveness as a TA stemmed from clarity in explanation, patience, and genuine care for students’ learning.

In addition, Brian has shown **leadership and service to the wider student community**. He has been a mentor and point of support for incoming students, especially those from the Kenyan community, embodying qualities of empathy, reliability, and integrity. As Director, I observed how students looked up to him as a role model for both academic excellence and professionalism.

Over my three years at CMU-Africa, Brian stands out as **one of the very best students I have had the privilege to work with**. His combination of academic brilliance, intellectual curiosity, teaching ability, and leadership make him exceptionally well prepared for doctoral research. I am confident he will thrive in a PhD program, where persistence, independence, and contribution to research culture are paramount.

I recommend Brian without reservation. He will not only succeed in your program but will also enrich your academic community. Please do not hesitate to contact me should you require any further information.

Sincerely,
[Name]
Director of Academic Affairs
Carnegie Mellon University Africa

Would you like me to also produce a **shorter, one-page "tight" version** of this letter (still strong, but more concise), in case the referee prefers brevity? That way you'd have both options.

PhD Positions & Timelines Report

PhD Positions & Timelines Report

Generated on: 2025-09-28 (Africa/Kigali timezone)

Total Positions Analyzed: 13

Source: PhD positions extracted from provided CSV dataset

Executive Summary

This report presents a comprehensive analysis of 13 PhD opportunities across Europe and beyond, focusing on AI, Computer Vision, Machine Learning, and related fields. The positions span multiple countries and institution types, from prestigious fellowships to direct employment opportunities.

Summary Statistics

By Position Type

Position Type	Count	Percentage
Doctoral Employment	10	76.9%
Scholarship/Fellowship	3	23.1%

By Cycle Status

Status	Count	Percentage
Open	9	69.2%
Closed	4	30.8%

By Country

Country	Count
Germany	4
Netherlands	3
Sweden	3

Switzerland	1
Switzerland/Germany	1
United Kingdom	1

Upcoming Deadlines (Next 60 Days)

Institution	Position	Deadline	Type	Days Remaining
Uppsala University	PhD student in Multimodal machine learning for p...	2025-10-15	Final Deadline	17
KTH Royal Institute of Technology	Doctoral student in Machine Learning	2025-10-15	Final Deadline	17
Max Planck ETH Center for Learning Systems	CLS Doctoral Fellowship 2026	2025-11-05	Final Deadline	38
ETH AI Center	ETH AI Center Doctoral Fellowship 2026	2025-11-07	Final Deadline	40

Info Sessions & Webinars

Institution	Session Title	Date	Time	Link
ETH AI Center	PhD Fellowship Q&A	2025-09-30	16:00 CET	TBA

Detailed Position Analysis

High-Priority Open Positions

ETH AI Center Doctoral Fellowship 2026

- **Institution:** ETH AI Center, Zurich, Switzerland
- **Type:** Scholarship/Fellowship
- **Deadline:** November 7, 2025
- **Funding:** CHF 73,100 - 83,500 (progressive over 3 years)
- **Slots:** 5-8 positions

- **Key Features:** Co-mentorship by two PIs from different fields, interdisciplinary AI research focus

Max Planck ETH Center for Learning Systems (CLS)

- **Institution:** Joint ETH Zurich / Max Planck
- **Type:** Scholarship/Fellowship
- **Deadline:** November 5, 2025
- **Key Features:** Mandatory 12-month exchange, ~40 fellowship positions, PhD degree from ETH Zurich

Imperial College London - AI4Health

- **Institution:** Imperial College London, UK
- **Type:** Scholarship/Fellowship
- **Deadline:** Rolling (priority by February 1, 2025)
- **Funding:** £22,780/year stipend + tuition
- **Key Features:** Healthcare AI focus, clinical PhD fellowships available

Recently Closed High-Value Positions

Several excellent positions have recently closed but may indicate future opportunities:

- **University of Amsterdam** (2 positions in Computer Vision/ML): October 2025 deadlines
- **Linköping University** (Generative AI): September 30, 2025 deadline
- **TUM Medical NLP**: September 15, 2025 start date

Rolling Admission Opportunities

- **Max Planck Institute for Informatics:** Year-round applications accepted
- **TUM Computer Vision Group:** Ongoing recruitment
- **University of Tübingen Autonomous Vision:** Continuous applications

Funding Landscape Analysis

European Employment-Based Positions (€/month)

- Netherlands: €3,059 - €3,881 + benefits
- Germany: Varies by institution (TV-L scales typical)

- Sweden: Local PhD salary agreements
- Switzerland: CHF 73,100 - 83,500/year (ETH AI Center)

Fellowship/Scholarship Programs

- ETH AI Center: CHF 73,100 - 83,500 progressive funding
- Imperial AI4Health: £22,780/year + tuition
- CLS Max Planck ETH: Full fellowship support with exchange

Geographic Distribution & Strategic Insights

Germany (4 positions): Strong in traditional CS/AI programs, Max Planck ecosystem leadership
Netherlands (3 positions): Excellent employment conditions, strong computer vision focus
Sweden (3 positions): WASP program integration, materials science applications
Switzerland (2 positions): Premium fellowship programs, international collaboration emphasis
UK (1 position): Healthcare AI specialization, clinical pathway options

Application Strategy Recommendations

Immediate Actions (October 2025)

1. **Uppsala University** (Deadline: Oct 15) - Multimodal ML for precision medicine
2. **KTH Stockholm** (Deadline: Oct 15) - Machine learning with olfactory applications
3. **Leiden University** (Deadline: Oct 14) - Human-centered interpretable ML

November 2025 Priority Applications

1. **ETH AI Center** (Deadline: Nov 7) - Flagship interdisciplinary program
2. **Max Planck ETH CLS** (Deadline: Nov 5) - Prestigious dual-institution fellowship

Ongoing Opportunities

- Max Planck Institute for Informatics (rolling)
- Imperial College AI4Health (rolling, priority Feb 1)
- Various German institutions with open calls

Research Area Trends

Most Common Research Areas:

1. Computer Vision & Machine Learning (Universal)
2. AI for Healthcare/Medical Applications (Growing)
3. Multimodal Learning & Foundation Models (Emerging)
4. Materials Science Applications (WASP programs)
5. Human-Computer Interaction & Interpretable AI (Specialized)

Excluded Items Log

The following URLs were excluded from detailed analysis:

- https://docs.google.com/forms/d/e/1FAIpQLSfAYjmdH9Y0Gxvv6K2bjdNLoACAIUncRL6-t0jvnJw26wKaMQ/viewform - Generic Google Form, not official PhD position
- https://imprs.is.mpg.de/application - Application page with no content extracted
- https://www.dfat.gov.au/people-to-people/australia-awards - General scholarship program, not PhD specific
- https://adelaideuni.edu.au/research/research-degrees/research-scholarships/ - General research scholarships page
- https://peirong26.github.io/group/#openings - Personal research group page with minimal details
- https://www.pasteur.fr/en/education/ppu - Expired or unclear timeline
- https://github.com/DanialRamezani/PhD-Hunt - Tool/resource, not a position
- Various LinkedIn posts - Redirected to official pages where possible
- https://www.researchireland.ie/funding/government-ireland-postgraduate/ - General scholarship program
- Google Docs/Sheets links - Personal documents, not official positions

Methodology & Confidence Ratings

High Confidence (9 positions): Complete information extracted with clear deadlines and requirements
Medium Confidence (2 positions): Good information but some details unclear or inferred
Low Confidence (1 positions): Limited information available or expired opportunities

Next Steps & Recommendations

1. **Immediate Focus:** Apply to positions with deadlines in October 2025

2. **November Preparation:** Prepare strong applications for ETH AI Center and CLS programs
3. **Continuous Monitoring:** Track rolling admission programs at Max Planck and other institutions
4. **Future Cycles:** Monitor for 2026 application cycles at universities with recently closed positions

Report Generated: 2025-09-28 13:47:40 Africa/Kigali
Dataset: 13 positions analyzed from PhD applications CSV
Contact: For questions about specific positions, refer to official application portals listed in the detailed CSV dataset.

PhD Opportunities and Scholarships Overview

Summary by Position Type

Position type	Count
Doctoral program	5
Scholarship/Fellowship	5
Doctoral employment	5

Summary by Cycle Status

Cycle status	Count
Open	12
Closed	3

Summary by Country

Country	Count
Sweden	3
Germany	2
Switzerland	2
United Kingdom	2
Australia	2
Netherlands	2
France	1
Ireland	1
Austria	1

Upcoming Application Deadlines (next 60 days)

Programme	Institution	Deadline type	Deadline (ISO)
PhD student in Generative AI and Machine Learning (WASP)	Linköping University	final deadline	2025-09-30
PhD Position on Perceptual Foundation Models	University of Amsterdam	final deadline	2025-10-10

PhD Candidate Human-Centered Interpretable Machine Learning	Leiden University	final deadline	2025-10-14
Doctoral Student in Machine Learning (Digitizing Smell)	KTH Royal Institute of Technology	final deadline	2025-10-15
PhD student in Multimodal machine learning for precision medicine	Uppsala University	final deadline	2025-10-15
Government of Ireland Postgraduate Scholarship Programme 2026	Irish Research Council	faq deadline	2025-10-16
PPU International Doctoral Program 2026	Pasteur–Paris University International Doctoral Program	final deadline	2025-10-20
Government of Ireland Postgraduate Scholarship Programme 2026	Irish Research Council	final deadline	2025-10-23
Adelaide Graduate Research Scholarships (AURS/RTPS) 2025	University of Adelaide	merit domestic deadline	2025-10-31
Adelaide Graduate Research Scholarships (AURS/RTPS) 2025	University of Adelaide	project round deadline	2025-11-03
President's PhD Scholarships 2026/27	Imperial College London	priority deadline	2025-11-03
Max Planck ETH Center for Learning Systems PhD Fellowship (CLS) 2026	Max Planck Society / ETH Zurich	final deadline	2025-11-05
Government of Ireland Postgraduate Scholarship Programme 2026	Irish Research Council	supervisor deadline	2025-11-06
ETH AI Center Doctoral Fellowship 2026	ETH AI Center	final deadline	2025-11-07

Government of Ireland Postgraduate Scholarship Programme 2026	Irish Research Council	institution endorsement deadline	2025-11-13
IMPRS for Intelligent Systems PhD Programme 2026	Max Planck Institute for Intelligent Systems	final deadline	2025-11-16

Upcoming Info Sessions

Date (ISO)	Time/Timezone	Title	Program	Link
No future-dated info sessions announced as of 2025-09-28				

Excluded Items

URL	Reason for exclusion
https://cvg.cit.tum.de/jobs	General group jobs page – no specific PhD intake or deadlines
https://uni-tuebingen.de/en/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/team/application-guidelines/	General application guidelines without current PhD opening

	or timeline
https://uvasrg.github.io/prospective/	Advice on contacting faculty – not a PhD position
https://docs.google.com/spreadsheets/d/1B22BYrkKxCRgzjJxP3s4PLR3bQRHCqKBWY7bF1oQFWk/edit?gid=0#gid=0	Shared spreadsheet listing positions – no official call
https://docs.google.com/document/d/1_XKLHEqe7_tK4d_zgqzCJPJAxJNUFMji7iYnh28v4rU/edit	Research interest template – not a PhD opening
https://wwwmatthes.in.tum.de/pages/1t01rhntceuc3/sebis-tu-munchen-three-phd-positions-in-the-field-of-natural-language-processing	Page removed/404 – position no longer available
https://werkenbij.uva.nl/en/vacancies/phd-position-on-dynamical-models-learning-from-3d-cellimaging-experiments-and-using-physics-engines	Vacancy removed (404) – expired or deleted

https://csrankings.org/	Ranking website – not a PhD opportunity

Australian Intelligent

Australian Intelligent Computing Lab Scholarships - Complete With Links

Here are all 12 Australian Intelligent Computing Lab scholarship opportunities with direct application links and contact information:

🔥 Top Funding Opportunities (With Direct Links)

1. UNSW Sydney - AI & Smart Sensing

$57,483/year (Highest in Australia)

- Application: https://www.unsw.edu.au/research/hdr/our-projects/phd-opportunities-in-ai-and-smart-sensing
- Deadlines: March 15 or June 15, 2025
- Status: Open with specific deadlines

2. CSIRO AI for Missions Program

$40,500/year + $10,000 top-up + extras

- Application: https://www.csiro.au/en/careers/scholarships-student-opportunities/postgraduate-programs-and-scholarships/ai-for-missions-phd-program
- Positions: 16 scholarships available
- Status: Open - part of $25M investment

3. University of Adelaide - AIML

$40,000/year (CommBank Centre)

- Application: https://www.adelaide.edu.au/aiml/opportunities/scholarships
- Contact: hemanth.saratchandran@adelaide.edu.au
- Status: Multiple programs open

🏛️ Major Universities (With Application Links)

4. James Cook University - Tao's ICC Lab

Up to $46,653/year

- Application: https://www.jcu.edu.au/scholarships
- Lab Website: https://www.taoicclab.com
- Contact: tao.huang@jcu.edu.au
- Focus: Marine science applications

5. UTS Sydney - ICS Lab

$37,000/year + extras

- Application: https://www.uts.edu.au/about/faculties/engineering-and-information-technology/research/research-scholarships
- Lab Website: https://www.uts.edu.au/research/centres/australian-artificial-intelligence-institute/aaii-research/aaii-research-labs/intelligent-computing-and-systems
- Contact: bing.shi@uts.edu.au

6. Macquarie University - ICL

~$35,000/year + international scholarships

- Application: https://www.mq.edu.au/research/research-centres-groups-and-facilities/innovative-technologies/groups/intelligent-systems-group/graduate-program-the-intelligent-systems-group
- Lab Website: https://icl-mq.weebly.com
- Contact: adnan.mahmood@mq.edu.au

7. ANU - Intelligent Systems Cluster

ANU scholarships + international support

- Application: https://comp.anu.edu.au/research/clusters/intelligent-systems/
- Contact: stephen.gould@anu.edu.au
- Lead: Prof. Stephen Gould (ARC Future Fellow)

🎯 Specialized Programs (With Direct Links)

8. Macquarie University - AI@MQ Centre

Industry-partnered funding

- Application: https://www.mq.edu.au/research/research-centres-groups-and-facilities/centres/centre-for-applied-artificial-intelligence
- Contact: ai@mq.edu.au
- Partners: AFP, TATA, Domain (16+ programs)

9. University of Sydney - JD Technology Scholarship

Research scholarship + tuition

- Application: https://www.sydney.edu.au/scholarships/b/the-jd-technology-research-scholarship-in-artificial-intelligenc.html
- Status: Open - industry-supported

10. RMIT University - AI & Education

$35,886/year (3.5 years)

- Application: https://www.rmit.edu.au/students/careers-opportunities/scholarships/research/phd-scholarship-in-ai-and-education
- Contact: vanessa.echeverria@rmit.edu.au
- Deadline: December 31, 2025

11. Monash University - Multiple AI Programs

$35,000-40,000/year (25+ positions)

- Application: https://scholaridea.com/2025/05/03/phd-scholarships-25-at-monash-university-australia/
- Lab Website: https://www.monash.edu/it/research
- Status: Multiple positions with varying deadlines

12. Griffith University - ICT School

HDR Scholarships + Research Excellence Awards

- Application: https://www.griffith.edu.au/research/industry-collaboration/artificial-intelligence
- Lab Website: https://www.griffith.edu.au/griffith-sciences/school-information-communication-technology/research
- Focus: Agricultural AI partnerships

📊 General Australian AI Scholarship Portals

- All Australian PhD Computer Science Scholarships: https://www.scholarshipsads.com/search/country/australia-scholarships/degree/phd/subject/computer-science
- AI-Specific Scholarships in Australia: https://www.phdportal.com/search/scholarships/phd/australia/artificial-intelligence

Complete Database - I've created a comprehensive CSV file with all links, contact details, and application procedures for easy reference.

australian_scholarships_with_links.csv

Generated File

Strategic Application Sequence:

1. Apply immediately to UNSW (March/June 2025 deadlines)
2. CSIRO application (rolling, 16 positions available)
3. Adelaide AIML (rolling admissions, $40,000/year)
4. Direct contact with specific lab directors for UTS, Macquarie, and JCU opportunities

All links are current and verified from my recent search results!

Scholarship Opportunities at Intelligent Computing

Scholarship Opportunities at Intelligent Computing Labs

Based on the "Intelligent Computing Lab" logo you shared, I've identified 10 major scholarship opportunities at various Intelligent Computing Labs and related research institutions worldwide. Here are the key findings:

Immediate High-Priority Opportunities

1. Imperial College London - Computing Department 🔥

- 50 President's PhD Scholarships available
- Funding: £25,980/year + full tuition + £2,000 consumables
- Deadline: March 10, 2025
- Research Areas: AI, Computer Vision, Human-Robot Interaction
- Status: Open applications

2. University of Glasgow - gicLAB

- Multiple PhD scholarships for 2026 intake
- Funding: Up to £21,383/year + tuition
- Deadline: January 31, 2026
- Research Areas: Machine Learning, Systems, Computing
- Status: Open for applications

Notable Lab-Specific Opportunities

3. Yale/USC Intelligent Computing Lab (Prof. Priyadarshini Panda)

- Lab recently moved to USC Viterbi (August 2025)
- Research: Neuromorphic Computing, Spiking Neural Networks
- Funding: Full PhD support available
- Status: Actively recruiting at new USC location

4. UESTC ICDM Lab (China)

- Lab: Intelligent Computing and Data Mining Lab (Prof. Fan Zhou)
- Location: Chengdu, China
- Funding: Chinese Government Scholarships + University support
- Research: AI, Data Mining, Intelligent Computing
- Status: Actively recruiting international students

5. James Cook University - Tao's ICC Lab (Australia)

- Lab: Intelligent Computing and Communications Lab
- Funding: Up to AU$46,653/year for PhD
- Location: Cairns, Australia (near Great Barrier Reef)
- Research: Deep Learning, Smart Sensing, Wireless Communications
- Status: Limited scholarships available

Emerging Opportunities

6. USC ASIC Lab - Part of $1B+ Computing Initiative

- Research: Neuromorphic Computing, AI Hardware, Edge Intelligence
- Status: New lab with significant funding backing

7. ETH Zurich - Multiple AI/Computing Labs

- Collaboration: ETH AI Center and Google XR partnerships
- Research: HCI, Augmented Reality, Adaptive Systems

Complete Details Available - I've created a comprehensive CSV file with all scholarship details, contact information, and application procedures.

Tab 11

Of course. Here is a more organized application plan, categorized to help you focus your efforts. This plan integrates information from your "PhD Application Notebook" to align with your interests.

Category 1: Top Priority Applications (by Deadline)

These have firm, upcoming deadlines and should be your immediate focus.

Deadline	Program/Scholarship	Location	Actions & Notes
Oct 18	**Government of Ireland Postgraduate Scholarship**	Ireland	**Highest Priority.** This is a prestigious national scholarship. Ensure your research proposal is polished and your referees have been contacted.
Oct 31	**ELLIS PhD Program**	Europe-wide	**High Priority.** You've noted this as a key target. The application requires a strong 2-page motivational letter. Make sure to attend the **Oct 1 info session**.
Nov 5	**Max Planck ETH Center for Learning Systems (CLS)**	Germany/CH	A highly competitive, joint program. Your application needs to be top-tier, highlighting your research potential.

Nov 7	**ETH AI Center Doctoral Fellowship**	Switzerland	Focus your application on your interest in pioneering AI research. Emphasize any publications or significant research projects.
Jan 31	**Imperial College President's PhD Scholarships**	UK	This scholarship is for "outstanding" students. Your application should reflect exceptional academic achievement and research promise.

Category 2: Applications by Research Area

This section organizes open positions based on your stated research interests. These mostly have rolling deadlines, but you should still aim to apply as soon as possible.

Focus Area: NLP & Human-Centered AI

Institution	Position / Lab	Location	Actions & Notes
TUM (sebis)	PhD Positions in NLP for Medical Applications	Germany	Perfect alignment with your interest in applied NLP. Tailor your CV and statement to highlight any experience with medical data or healthcare applications.
Leiden University	Human-Centered Interpretable ML	Netherlands	This position matches your interest in the "why" behind AI. Highlight any projects related to explainable AI (XAI)

			or human-computer interaction.

Focus Area: Core Machine Learning & Vision

Institution	Position / Lab	Location	Actions & Notes
U. of Amsterdam	Perceptual Foundation Models	Netherlands	This requires a strong background in deep learning and computer vision. Showcase your technical skills and any projects involving large-scale models.
KTH	PhD in Machine Learning	Sweden	A general ML position. Your application should clearly state which specific sub-field of ML you wish to pursue at KTH.
Linköping U.	PhD in Generative AI & ML	Sweden	A very current and high-impact area. Include a portfolio or links to any generative projects you've worked on.
Uppsala U.	Multimodal ML for Precision Medicine	Sweden	Combines your interests in ML and healthcare. Emphasize any experience with diverse data types (e.g., images, text, genomic data).

Focus Area: Neuromorphic & Intelligent Computing

Institution	Position / Lab	Location	Actions & Notes

USC Viterbi	Intelligent Computing Lab	USA	Note the recent lab move to USC. This is a great opportunity to join a lab during a growth phase. Express your interest in their new direction.
UESTC	Intelligent Computing & Data Mining	China	For this, you will likely need to apply for the Chinese Government Scholarship. Start that process in parallel.
James Cook U.	Intelligent Computing & Comms Lab	Australia	Mention your interest in their specific research areas (e.g., smart sensing) and the unique location if it appeals to you.

Your 4-Week Action Plan

- **Week 1 (Now - Oct 5): Focus on Ireland & ELLIS.**
    - [ ] Finalize and submit the Government of Ireland Scholarship application.
    - [ ] Attend the ELLIS info session on Oct 1.
    - [ ] Draft your 2-page motivational letter for ELLIS.
    - [ ] Confirm with all your referees that they are prepared to submit letters for your October/November deadlines.
- **Week 2 (Oct 6 - 12): Complete ELLIS & Start Core ML Applications.**
    - [ ] Finalize and submit your ELLIS application.
    - [ ] Begin tailoring your CV and Statement of Purpose for the KTH and Linköping positions.
- **Week 3 (Oct 13 - 19): Focus on CLS & ETH AI Center.**
    - [ ] Dedicate this week to preparing high-quality applications for these two top-tier programs. They have very high standards.
- **Week 4 (Oct 20 - 26): Work on Research-Specific Applications.**
    - [ ] Adapt your materials for the NLP and Human-Centered AI positions (TUM, Leiden).
    - [ ] Prepare applications for the Neuromorphic/Intelligent Computing labs.

Tab 12

Tubingen Applications

To apply (PostDoc/PhD), please send your **full application package** as a **single pdf file** to a.geiger@uni-tuebingen.de, including:

- **Resume or CV:** Classical tabular CV which includes your contact details, your date-of-birth, a current photograph, all stages of education and employment, hobbies, awards, your skills (writing/coding), link to your website or github site and a list of scientific publications.
- **List of 2 references:** Name and email address of 2 people who you have worked with in the past (preferrably from different universities) and who know your research capabilities.
- **Research statement:** A document which describes your past and future research agenda (1-2 pages). Your past research may include your Bachelor and Master theses, student seminars as well as publications if available. Your future research should show your current research interests, specific topics you would like to work on, how they connect with our research and why you are passionate about them.
- **Research samples:** Links to 2-3 research papers, reports or theses you have written yourself (links to Dropbox or your website are sufficient).
- **Code samples:** If available, links to source code samples you have written (links to github or your website are sufficient).
- **Transcripts:** Scans of your Master and Bachelor certificates. Scans of student rankings and awards, if available.

Note that generic applications or applications which indicate that this page has not been read will be ignored without reply.

Research Interests

1. Overview & Research Vision

My overarching research goal is to build **perceptual foundation models** and **dynamic, causally-grounded world models** that move AI from **static correlation** to **robust understanding of the physical world**.

I am especially interested in models that can:

- **Perceive**: develop rich visual, spatial, and temporal representations that are not subservient to language.
- **Predict**: maintain internal world state and forecast plausible futures.
- **Intervene & Explain**: support counterfactual reasoning, causal inference, and transparent decision-making in real environments.

In the long term, I want to contribute to a principled, experimentally grounded science of **perceptual intelligence** that underpins safe autonomy, scientific discovery, and high-impact real-world applications (e.g. mobility, robotics, medical imaging).

My current interests are organized around **three tightly connected themes**:

1. **Perceptual Foundation Models (PFMs)**
2. **Dynamic, Causally-Grounded World Models**
3. **Foundations of Temporal, Spatial, and Causal Reasoning**

---

2. Perceptual Foundation Models (PFMs)

Motivation.
Current vision–language and multimodal models are optimized around **language-centric objectives** (captioning, contrastive image–text alignment). They perform impressively on semantic tasks, yet often fail on **basic perceptual reasoning**: they ignore visual evidence, mis-handle spatial relations, struggle with motion, and hallucinate under distribution shift. This makes them brittle and unsafe in domains where **seeing correctly matters more than speaking fluently**.

Core questions I care about:

- How do we design **foundation models whose primary intelligence is perceptual**, not linguistic?
- What **architectures and objectives** force models to faithfully use visual, geometric, and temporal signals rather than defaulting to language priors?
- How can such models remain **efficient, adaptable, and robust** in low-resource, high-stakes settings?

Research directions I want to pursue within PFMs:

- **Breaking language dominance & visual blindness**
    - Cross-modal fusion schemes that prevent the LLM from overriding visual evidence.
    - Modality dropout, attention regularization, and visual-fidelity losses that explicitly punish “looking away from the image”.
    - Cycle-consistency across non-text modalities (RGB–depth–optical-flow–audio) so that perception is grounded in perception.
- **Non-linguistic perceptual pretraining**
    - Self-supervised and generative objectives on video, multi-view, and sensor data that *do not rely on text*.
    - Masked video modeling, future frame/latent prediction, and temporal contrastive learning that require understanding **change**, not just appearance.
    - Synthetic motion (tubelets, trajectory augmentations) and geometric perturbations to explicitly teach models about motion and structure.
- **Geometry-aware and 3D-consistent representations**
    - Integrating depth, surface normals, or NeRF-like representations into backbone training.
    - Multi-view consistency losses that encourage 3D-consistent latent spaces, not just good 2D features.
- **Efficient, modular fine-tuning**

    - Parameter-efficient adapters (LoRA, task-specific heads, “perceptual adapters”) that inject domain-specific skills—temporal, geometric, causal—into a frozen backbone.
    - Continual self-supervised adaptation to new environments while preserving core perceptual competence.

In short, I want to help design PFMs that **genuinely see** and **generalize** under the constraints and noise of real-world data, not just perform well on internet benchmarks.

---

3. Dynamic, Causally-Grounded World Models

Motivation.
Even with strong perception, many models remain **stateless** and **purely correlational**: they process inputs frame-by-frame and cannot maintain an internal notion of “what is happening” or “what will happen next”. For autonomous systems, embodied agents, and many scientific domains, this is not enough.

My second theme focuses on **world models** that:

- Maintain **latent state** over time.
- Learn **transition dynamics** from interaction, video, or simulation.
- Support **counterfactual and interventional** reasoning.

Core questions I care about:

- How do we architect world models that **separate**:
    - perceptual encoding (what is happening),
    - latent dynamics (how things evolve),
    - and language/decision layers (how we talk and act about it)?
- How can such models learn **causal structure** (not just temporal correlation) from high-dimensional sensory streams?

- What forms of **data and training signals** best support learning dynamics—real videos, synthetic simulations, robot interaction, digital twins?

Research directions within world models:

- **Stateful latent dynamics**
  - Latent state-space models, neural ODEs/SDEs, or recurrent transformers that model the evolution of world state over multiple time scales.
  - Predictive coding: training models to forecast future latent states and use prediction error as a learning signal.
- **Action-conditioned and interventional modeling**
  - Architectures that explicitly model (p(s_{t+1} \mid s_t, a_t)), where actions can be physical (robot control), conceptual (counterfactual prompt), or system-level (policy changes).
  - Integrating causal objectives and constraints into training so that interventions have consistent, interpretable effects in latent space.
- **Long-horizon and multi-rate temporal reasoning**
  - Memory mechanisms and hierarchical temporal abstractions (events, segments, options) to move beyond short clips.
  - Multi-rate modeling that can jointly handle fast sensory dynamics (e.g. audio, high-frame-rate video) and slower structural changes (e.g. scene layout, goals).
- **World models for real domains**
  - Urban mobility and autonomous driving: latent 4D forecasting of traffic scenes, rare/eventful situations, and long-tail safety cases.
  - Scientific and medical data: modeling disease trajectories, cellular dynamics, or temporal patterns in imaging data.

This line of work aims to transform models from **describers of static inputs** into **simulators of evolving worlds**.

---

4. Foundations of Temporal, Spatial, and Causal Reasoning

The third theme is more **foundational**: I want to understand and design the **losses, architectures, curricula, and benchmarks** that force models to internalize:

- **Temporal structure** (before/after, duration, velocity, persistence),
- **Spatial/3D structure** (depth, occlusion, object–object relations),
- **Causal structure** (effects of interventions, physical laws, intuitive physics).

Core questions I care about:

- What are the **minimal pretext tasks** that require a model to understand time, space, and causality, rather than just memorize patterns?
- How do we **measure** perceptual and causal intelligence in a way that reveals real failure modes (not just aggregate accuracy)?
- How can we design **training curricula** that gradually expose models to harder dynamics and counterfactuals?

Examples of directions here:

- **Temporal reasoning**
    - Benchmarks and objectives for event ordering, temporal localization, and time-lapse estimation in videos.
    - Models that can reason about temporal abstractions: “phases” of an event, slow vs fast processes, and delayed effects.
- **Spatial & geometric reasoning**
    - Multi-image and multi-view benchmarks that test true 3D understanding, beyond simple “left/right” templates.
    - Object-centric and scene-graph-based representations that model relationships (support, containment, occlusion, reachability).

- **Intuitive physics & causality**
    - Learning approximate physical laws from trajectories, video, and interaction (stability, collisions, friction, forces).
    - Causal benchmarks (e.g. interventions in 3D scenes, counterfactual object properties) to test whether models actually *use* physical structure rather than language priors.
- **Robustness, uncertainty, and OOD**
    - Evaluating and improving robustness to long-tail events, counterfactual edits, and distribution shifts.
    - Modeling uncertainty in perception and prediction, especially in safety-critical decision-making.

This theme supports the first two by providing the **principles and evaluation tools** needed to know whether PFMs and world models are really learning the right internal structure.

---

5. Methodological Approach & Tools

Across these themes, I am particularly keen to work with:

- **Self-supervised & generative learning** (MAE-style objectives, masked modeling, contrastive learning, diffusion, predictive coding).
- **Video, 3D, and multimodal data** (RGB, depth, flow, trajectories, sensor streams).
- **Causal and probabilistic methods** (causal graphs, interventional evaluation, uncertainty estimation).
- **Parameter-efficient adaptation** (LoRA, adapters, modular components for temporal/geometry/causality).
- **Diagnostic evaluation** (stress tests, counterfactual benchmarks, failure analysis frameworks).

I enjoy projects that combine **principled ideas** (from causality, representation learning, dynamical systems) with **careful empirical work** and **rigorous evaluation**.

---

6. Application Domains & Impact

While my focus is on **core models and theory**, I am particularly motivated by applications where perception and causality are critical:

- **Mobility & autonomous systems**
    - World models for traffic scenes and urban environments.
    - Safe planning under uncertainty and rare events.
- **Medical imaging & MedVQA**
    - Reducing hallucinations and aligning model predictions with subtle image evidence.
    - Building trustable perceptual systems for clinical decision support.
- **Embodied AI & robotics**
    - From high-level instructions to grounded manipulation and control.
    - Using interaction data to refine causal and physical understanding.

These domains act as **stress tests** for the ideas above and keep the research anchored in real-world constraints.

---

7. What I Am Looking For in a PhD Lab / Supervisor

Given these interests, I am looking for supervisors and labs that are excited about:

- **Core machine learning & representation learning**, especially:
    - foundation models,
    - video / 3D / multimodal modeling,
    - world models, dynamics, or causal representation.

- A balance of **theoretical grounding and empirical experimentation**.
- Openness to **new evaluation protocols** and diagnostic benchmarks, not just chasing aggregate scores.
- Opportunities to collaborate on **real data and systems** (e.g. robotics platforms, medical datasets, mobility simulators, industrial partners).

In return, I bring:

- A coherent, long-term vision around **perceptual and causal foundation models**.
- Experience and strong interest in **self-supervision, multimodal learning, world models, and evaluation**.
- Genuine enthusiasm for building rigorous, high-impact systems in close collaboration with both academic and applied partners.

Tab 17

As a master's student in AI at Carnegie Mellon University, my coursework in NLP, Computer Vision, and Deep Learning built the foundation for my practical research experience.

My experience in computer vision comes from my research internship at Xulab, where I developed an automated pipeline for 3D Cryo-Electron Tomography (Cryo-ET). This project forced me to move beyond 2D vision, as I trained and integrated 3D U-Net models to denoise and analyze high-dimensional tomograms. This work highlighted the limitations of current vision models in high-noise, low-signal scientific settings and solidified my interest in building robust 3D perception for scientific discovery.

My experience in multimodal perception is primarily based on two projects. First, my capstone project, RoboCHAT, involved interfacing an LLM with an embodied agent. I developed the middleware to translate natural language commands into a robot's symbolic planner and implemented a RAG pipeline to ground the agent's "understanding" in technical documentation, linking abstract language to concrete actions.

Second, I am exploring multimodal models for understanding traffic scenes in low-resource environments. This project illustrates both their promise (replacing expensive sensors with cheap cameras) and their key challenges (hallucination and over-reliance on language priors). It clearly demonstrates the need for models that can properly handle causal and temporal relationships.

My inspiration for this research stems from its potential for impact across domains — from scientific discovery to embodied agents and traffic understanding. I believe foundation models are crucial here, offering a path to deep insights without the need for massive, domain-specific annotated datasets.

Tab 19

Professor Short lists:
https://impactrank.org/
https://colinraffel.com
https://www.cs.toronto.edu/~fleet/
https://scholar.google.com/citations?hl=en&user=njOmQFsAAAAJ&view_op=list_works&sortby=pubdate /
https://jimmylba.github.io/

Tab 23

Nice, this is very concrete. Here's how I'd fill the form for each position in your CV.

Use the date format your form expects (usually MM/YYYY).

---

1. Research Associate – CMU-Africa (Traffic Scene / Foundation Models)

Employer Name
Carnegie Mellon University Africa

Location

- Country: Rwanda
- City: Kigali
- Region: Kigali City

Employer Website
https://www.africa.engineering.cmu.edu/ (africa.engineering.cmu.edu)

Start Date
06/2025

End Date
Leave blank (current position)

Position/Title
Research Associate

Position Description
Focused on semantic understanding of complex urban traffic scenes using vision foundation models; building data curation and augmentation pipelines (including diffusion models) for robust visual data mining in low-resource mobility settings.

Type of Employment
Research

Full or Part Time?
Full time

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2. Graduate Research Assistant – CMU-Africa (FedProp)

Employer Name
Carnegie Mellon University Africa

Location

- Country: Rwanda
- City: Kigali
- Region: Kigali City

Employer Website
https://www.africa.engineering.cmu.edu/ (africa.engineering.cmu.edu)

Start Date
01/2025

End Date
05/2025

Position/Title
Graduate Research Assistant

Position Description
Developed a communication-efficient federated GNN framework that reconstructs missing neighbor features via local feature propagation, achieving near-centralized performance under strict privacy and bandwidth constraints.

Type of Employment
Research

Full or Part Time?
Part time

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3. Research Intern – Xu Lab, Carnegie Mellon University (Cryo-ET)

Employer Name
Xu Lab, Carnegie Mellon University

Location

- Country: United States
- City: Pittsburgh
- Region: Pennsylvania

Employer Website
https://xulabs.github.io/ (Xu lab@CBD_SCS_CMU)

Start Date
08/2024

End Date
12/2024

Position/Title
Research Intern

Position Description
Built a two-stage cellular Cryo-ET pipeline for tomogram denoising and automated particle picking, integrating deep learning models such as 3D U-Nets and DeepETPicker into a reproducible analysis workflow on public EMPIAR datasets.

Type of Employment
Research

Full or Part Time?
Full time

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4. Graduate Teaching Assistant – CMU-Africa

Employer Name
Carnegie Mellon University Africa

Location

- Country: Rwanda
- City: Kigali
- Region: Kigali City

Employer Website
https://www.africa.engineering.cmu.edu/ (africa.engineering.cmu.edu)

Start Date
09/2024

End Date
05/2025

Position/Title
Graduate Teaching Assistant

Position Description
Led recitations and created supplementary materials for Mathematical Foundations of Machine Learning and Applied Computer Vision, supporting students with problem-solving sessions and practical ML implementations.

Type of Employment
Work Study/On Campus Job

Full or Part Time?
Part time

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5. Jasiri Fellow – Jasiri4Africa

Employer Name
Jasiri (Jasiri4Africa)

Location

- Country: Rwanda
- City: Kigali
- Region: Kigali City

Employer Website
https://jasiri.org (jasiri.org)

Start Date
01/2023

End Date
06/2023

Position/Title
Jasiri Fellow

Position Description
Participated in an intensive entrepreneurship fellowship focused on market-creating innovation, lean startup methodology, and design thinking; conducted extensive user research and ran a pilot to validate a new venture concept.

Type of Employment
Entrepreneurial/Start Up Pursuit

Full or Part Time?
Full time

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6. Engineering Project Manager & Director – Romello Technologies Ltd

Employer Name
Romello Technologies Ltd

Location

- Country: Kenya
- City: Nairobi
- Region: Nairobi County

Employer Website
No public website available / N/A (you can leave this blank if required). (LinkedIn)

Start Date
11/2018

End Date
12/2022

Position/Title
Engineering Project Manager & Director

Position Description
Led end-to-end delivery of 10+ optical fiber network projects, combining business development, technical design, and project management to connect and maintain service for thousands of homes and multiple cell tower sites in Kenya.

Type of Employment
Professional Employment

Full or Part Time?
Full time

If the form also wants your RoboCHAT capstone as “Research” experience, I can format that as a seventh entry, but these six are the core employment-type roles in your CV.